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Reviewed and approved* by the following:

Some Person
Professor of Something
Thesis Supervisor

Some Other Person
Professor of Something Else
Some Other Position
Honors Adviser

*Signatures are on file in the Schreyer Honors College.

Abstract

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Chapter 1

Introduction and Motivation

1.1 Introduction

In Super Smash Bros. Melee (or colloquially known as just “Melee”) for the Nintendo GameCube, there are 26 characters to choose from, including fan-favorites like Mario and Luigi, to less known characters like Ness and Ice Climbers. In competitive Melee, each player has four lives, also known as stocks. A player wins by depleting all of their opponents stocks before losing theirs.

Typically, in competitive games or sports, each individual or team has equal access to the same moves and techniques. For example, in chess, each player has the same number and types of pieces. In game theory, this is an example of a symmetric game, in which the players have the same strategy set [1] p. 186. However, in fighting games, each character has a unique moveset that gives them strengths and weaknesses. This is an example of an asymmetric game, in which each character has a unique strategy and set of options that can be employed. Certain characters have swords or projectiles, allowing them to attack opponents from a distance. Some characters are fast, but easy to knock off the stage, while others are slow, but more resilient to being defeated.

Additionally, Melee is unique from other fighting games, in that it is what is known as a platform fighting game. Typically, fighting games are played on a two-dimensional plane with no frills. The goal is to deplete the health of your opponent. In Melee, the goal is to knock your opponent off of the stage to deplete their stocks. Each stage has unique properties that can affect how certain characters play. Characteristics such as platforms, stage transformations, and other interactable elements affect how players approach matchups and can directly affect gameplay. It is agreed upon in the community that certain stages are better for certain characters due to the tools available to the character.

Because of the asymmetric nature of Melee’s characters, it is natural to assume that some characters perform better against others due to having a superior toolkit in a given matchup. To express how strong a character is, players will create a “matchup chart” or “matchup spread” for the character. This is a measurement of a character’s probability of winning against other characters in the roster. When designing a matchup chart, it is typically assumed that the players of each character are of equal skill level. Matchup charts are also typically designed based on the perception of how character perform against one another, given their respective toolkits and performance on certain stages. There is typically little to no formal statistical analysis when designing these spreads.

While some matchup charts are expressed using $\pm 1, 2, \dots$, this is not as descriptive as to how favorable or unfavorable a matchup may be. We can also use explicit probabilities when describing a matchup. Formally, let us define a matchup of character A against character B. When discussing this matchup, we would say “Character A’s matchup against Character B is x/y ,” where x is the probability that character A would win, y is the probability that character B would win, and $x + y = 100$. For example, an even matchup would be designated as “50/50”. A slightly favorable matchup would be “55/45”, while a heavily unfavorable matchup may be expressed as “35/65”.

1.2 Motivation

A tier list is an ordered list of characters, where each character is ranked based on how strong they are relative to the rest of the cast of characters. Tier lists are often subjective, and they will change over time as the strategies for each character develops over time.

One of the most prominent matchups in competitive melee is Fox vs. Marth. Both of these

characters have sat near the top of almost every competitive tier list for over two decades. Both characters have been extremely successful, and many players have piloted the characters to major tournament victories. They each have strong toolkits, and the ability to perform well against any character in the game. However, there has been debate in the Melee community about how Fox and Marth perform against each other.

Most players would argue that the matchup is even. Each character has tools in their moveset to exploit the other character's weaknesses. However, top Melee players have expressed different opinions. The most infamous claim was made by the player William "Leffen" Hjelte. Leffen, a Fox player, argued that Marth is favored in the matchup 60/40, and it became somewhat of a joke in the Melee community.

I thought that this debate would be interesting to approach from a Bayesian lens. Given expert prior beliefs on the matchup, we could use results from individual matches to determine a more statistically sound matchup spread for a character. Furthermore, Melee has many characters that get less usage due to being considered weaker than most choices. Thus, there is little data for some of these more obscure matchups. Bayesian methods thrive in this environment, allowing us to define such matchup spreads for less-used characters.

Chapter 2

Data Collection

2.1 Slippi

Melee was originally released in 2001, with limited features. [Slippi](#), released in June 2020, is a project designed to bring features of modern-day e-sports to Melee. These include portable replay files, complex gameplay stats, online matchmaking with improved online netcode, and more [2]. Not only did this make Melee more accessible through the introduction of online play, a feature missing from the original game, but Slippi also ushered in an era of data analytics surrounding Melee. Using Slippi game data, it is possible to easily analyze match data without having to manually record match results.

For the data used in this project, it was sourced from [Chart.slp](#), a platform that hosted a large corpus of Slippi match data. Although it is no longer maintained, it is possible to download a backup of the database, containing data for 5,594,571 different matches. This provides sufficient data for Bayesian analysis across nearly every matchup in Melee.

To get this data into a usable format, I developed a few Python scripts to groom the data. First, for this analysis, we only care about matchup data. The information in the database file used for match replay is not needed and consumes a large amount of space. We start by parsing just match data, such as characters, stage, winner, etc.

After parsing the file for just match data, another script is used to exclude incomplete matches. Slippi allows players to disconnect from a match early, which does not result in an explicit winner. Thus, these matches are filtered out. After this filter is applied, other scripts can be run to get matches for two specific characters, and data about the stage and winner can be exported into a CSV format for analysis.

Chapter 3

Modeling the Marth-Fox Matchup

3.1 Model 1: A Simple Binomial-Beta Conjugate

For the first model, we will approach it completely analytically. We will model the win rate of a character in a matchup, without factoring in stage choice. Let θ represent the probability of a character winning a given matchup. Let the likelihood of our model be represented as a binomial distribution $x_i|\theta \sim \text{Bin}(n, \theta)$, where n represents the number of matches in the sample. A win for the character would be recorded as a 1 (success), while the loss would be recorded as a 0 (failure). We can specify a Beta prior generally as $\theta \sim \text{Be}(\alpha, \beta)$. A Beta prior is flexible, as it allows us to experiment with different initial beliefs for a given matchup.

The Beta prior can form a conjugate distribution with the posterior with respect to the Binomial likelihood. This enables us to find a fully closed-form posterior. Using existing conjugacy tables [3] p. 341. This gives the posterior $\theta|X \sim \text{Be}(\alpha + x, \beta + n - x)$. The mean of the posterior, $\frac{\alpha+x}{(\alpha+x)+(\beta+n-x)}$, gives us the expected win rate for a character in a given matchup.

For the Marth-Fox matchup, assume that we are modeling Marth's win rate against Fox. Below are graphs representing the posterior matchup distribution, given different priors. Although this is a good exercise and a simple way to use Bayesian methods to better model matchups in Melee, there are several components not included in this initial model. First, as previously explained, stages in Melee often make a difference in a matchup. In the Marth-Fox matchup, it is argued that Final Destination is a strong choice for Marth due to him gaining access to techniques not available on stages with platforms. Thus, we are interested in also understanding character matchups by stage. Second, it is likely that not all of the observed matches were against evenly matched players. We may want to design a model that factors in player skill when trying to determine the true matchup spread.

3.2 Model 2: Marth-Fox Matchup by Stage

In competitive Melee, there are 6 tournament-legal stages. These are Battlefield, Final Destination, Dream Land, Fountain of Dreams, Pokemon Stadium, and Yoshi's Story. We are interested in understanding how the Marth-Fox matchup varies by stage. We can utilize the same Binomial-Beta model as previously used, but we instead will separate the data by stage. Again, assume we are modeling Marth's win rate against Fox. Below are graphs representing the win rate of Marth based on the stage chosen.

3.3 Model 3: Factoring in Elo

In newer versions of Slippi, a ranked mode was added. In this mode, players are given a rating. A higher number is indicative of a more skilled player, and vice versa. We can use this information as a factor in modeling the matchups in Melee.

Bibliography

- [1] Stephen Schecter and Herbert Gintis. *Game Theory in Action: An Introduction to Classical and Evolutionary Models*. Princeton University Press, 2016.
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- [3] Brani Vidakovic. *Engineering biostatistics: an introduction using MATLAB and WinBUGS*. Wiley, Hoboken, Nj, 2017.